

Variable Selection, [George and McCulloch, 1993¹](#)) or with other hypotheses that have a continuous prior distribution close to zero (e.g., the sceptical prior proposed by [Pawel and Held, in press](#)). As far as evidence from the marginal likelihood is concerned, the results below show that these methods are inconsistent.

We end with a brief discussion on how a consistent method for hypothesis testing can be obtained without fully committing to a point-null hypothesis.

2 The Peri-Null Correction Factor

Consider the three hypotheses discussed earlier: the point-null hypothesis $\mathcal{H}_0$ fixes the test-relevant parameter to a fixed value (e.g., $\delta = 0$); the peri-null hypothesis $\mathcal{H}_{\tilde{0}}$ assigns the test-relevant parameter a distribution that is tightly centered around the value of interest (e.g., $\delta \sim \pi(\delta | \mathcal{H}_{\tilde{0}}) = \mathcal{N}(0, \kappa_0^2)$ with κ_0 small); and the alternative hypothesis $\mathcal{H}_1$ assigns the test-relevant parameter a relatively wide prior distribution, $\delta \sim \pi(\delta | \mathcal{H}_1)$. The Bayes factor of interest is between $\mathcal{H}_1$ and $\mathcal{H}_{\tilde{0}}$, which can be expressed as the product of two Bayes factors involving $\mathcal{H}_0$:

$$\underbrace{\frac{p(y^n | \mathcal{H}_1)}{p(y^n | \mathcal{H}_{\tilde{0}})}}_{\text{Peri-null BF}_{1\tilde{0}}(y^n)} = \underbrace{\frac{p(y^n | \mathcal{H}_1)}{p(y^n | \mathcal{H}_0)}}_{\text{Point-null BF}_{10}(y^n)} \times \underbrace{\frac{p(y^n | \mathcal{H}_0)}{p(y^n | \mathcal{H}_{\tilde{0}})}}_{\text{Correction factor BF}_{0\tilde{0}}(y^n)}. \quad (3)$$

In words, the Bayes factor for the alternative hypothesis against the peri-null hypothesis equals the Bayes factor for the alternative hypothesis against the point-null hypothesis, multiplied by a correction factor (cf. [Kass and Vaidyanathan, 1992](#), [Kass and Raftery, 1995](#), [Morey and Rouder, 2011](#), p. 411). This correction factor quantifies the extent to which the point-null hypothesis outpredicts the peri-null hypothesis. With data sets of moderate size, and κ_0 small, the peri-null and point-null hypotheses will make similar predictions, and consequently the correction factor will be close to 1. In such cases, the point-null can indeed be considered a mathematically convenient approximation to the peri-null.

2.1 Example

Consider the hypothesis that “more desired objects are seen as closer” ([Balcetis and Dunning, 2011](#)). In the authors’ Study I, 90 participants had to estimate their distance to a bottle of water. Immediately prior to this task, 47 ‘thirsty’ participants had consumed a serving of pretzels, whereas 43 ‘quenched’ participants had drank as much as they wanted from four 8-oz glasses of water. In line with

¹ “A similar setup in this context was considered by Mitchell and Beauchamp (1988), who instead used “spike and slab” mixtures. An important distinction of our approach is that we do not put a probability mass on $\beta_i = 0$.” ([George and McCulloch, 1993](#), p. 883).

the authors’ predictions, “Thirsty participants perceived the water bottle as closer ($M = 25.1$ in., $SD = 7.3$) than quenched participants did ($M = 28.0$ in., $SD = 6.2$)” (Balcetis and Dunning, 2011, p. 148), with $t = 2.00$ and $p = .049$. A Bayesian point-null t -test concerning the test-relevant parameter δ may contrast $\mathcal{H}_0 : \delta = 0$ versus $\mathcal{H}_1 : \delta \in \mathbb{R}$ with a Cauchy distribution with location parameter 0 and scale κ_1 , the common default value $\kappa_1 = 1/\sqrt{2}$ (Morey and Rouder, 2018). The resulting point-null Bayes factor is $BF_{10} = 1.259$, a smidgen of evidence in favor of $\mathcal{H}_1$. We may also compute a peri-null correction factor by contrasting $\mathcal{H}_0 : \delta = 0$ against $\mathcal{H}_{\tilde{0}} : \delta \sim \mathcal{N}(0, \kappa_0^2)$, with $\kappa_0 = 0.01$, say. The resulting peri-null correction factor² is $BF_{0\tilde{0}} = 0.997$, which means that, practically, it does not matter if the point-null or the peri-null is tested. With a larger value of $\kappa_0 = 0.05$, we have $BF_{0\tilde{0}} = 0.927$, thus, a peri-null Bayes factor of $BF_{1\tilde{0}} = 1.167$. The change from $BF_{10} = 1.259$ to $BF_{1\tilde{0}} = 1.167$ is utterly inconsequential.

The difference between the peri-null and point-null Bayes factor remains inconsequential for larger values of t . When we change $t = 2.00$ to $t = 4.00$, the point-null Bayes factor equals $BF_{10} = 174$, which according to Jeffreys’s classification of evidence (e.g., Jeffreys, 1961, Appendix B) is considered compelling evidence for $\mathcal{H}_1$. With $\kappa_0 = 0.01$, the peri-null correction factor equals $BF_{0\tilde{0}} = 0.986$ and consequently a peri-null Bayes factor equals of about 172 in favor of $\mathcal{H}_1$ over $\mathcal{H}_{\tilde{0}}$. With $\kappa_0 = 0.05$, the peri-null correction factor equals $BF_{0\tilde{0}} = 0.713$ and $BF_{1\tilde{0}} \approx 124$. In absolute numbers, the change from 174 to 124 may appear considerable, but with equal prior model probabilities this translates to a modest difference in posterior probabilities: $P(\mathcal{H}_1 | y^n) = 174/175 \approx 0.994$ versus $124/125 = 0.992$.

The peri-null correction factor does become influential when sample size is large. As we prove in the next section, the peri-null Bayes factor is inconsistent and converges to the ratio of prior ordinates under $\mathcal{H}_1$ and $\mathcal{H}_{\tilde{0}}$ at the maximum likelihood estimate.

3 The Peri-Null Bayes Factor is Inconsistent

Historically, the main motivation for the development of the Bayes factor was the desire to be able to obtain arbitrarily large evidence for a general law: “We are looking for a system that will in suitable cases attach probabilities near 1 to a law.” (Jeffreys, 1977, p. 88; see also Etz and Wagenmakers, 2017; Ly et al., 2020; Wrinch and Jeffreys, 1921).

Statistically, this desideratum means that we want Bayes factors to be consistent, which implies that, as sample size increases, (i) $BF_{10}(Y^n)$ tends to zero when the data are generated under the null model, whereas (ii) $BF_{01}(Y^n)$ tends to zero when the data are generated under the alternative model $\mathcal{H}_1$,

² Calculated using the Summary Stats module in JASP, (e.g., Ly et al., 2018, jasp-stats.org), and based on Gronau et al. (2020).

that is,

$$\text{BF}_{10}(Y^n) \xrightarrow{\mathbb{P}_\theta} 0 \text{ if } \mathbb{P}_\theta \in \mathcal{H}_0, \text{ and } \text{BF}_{01}(Y^n) \xrightarrow{\mathbb{P}_\theta} 0 \text{ if } \mathbb{P}_\theta \in \mathcal{H}_1. \quad (4)$$

Thus, regardless of the chosen prior model probabilities $P(\mathcal{H}_0), P(\mathcal{H}_1) \in (0, 1)$,

$$P(\mathcal{H}_0 | Y^n) \xrightarrow{\mathbb{P}_\theta} 1 \text{ if } \mathbb{P}_\theta \in \mathcal{H}_0, \text{ and } P(\mathcal{H}_1 | Y^n) \xrightarrow{\mathbb{P}_\theta} 1 \text{ if } \mathbb{P}_\theta \in \mathcal{H}_1, \quad (5)$$

where $\mathbb{P}_\theta$ refers to the data generating distribution, here, $Y_i \stackrel{\text{iid}}{\sim} \mathbb{P}_\theta$, and where $X_n \xrightarrow{\mathbb{P}_\theta} X$ denotes convergence in probability, that is, $\lim_{n \rightarrow \infty} \mathbb{P}_\theta(|X_n - X| > \epsilon) = 0$ as usual.

Below we prove that the peri-null Bayes factor $\text{BF}_{10}(Y^n)$ is inconsistent (cf. suggestions by [Jeffreys, 1961](#), p. 367; [Jeffreys, 1973](#), p. 39, Eq. 2; and the statements by [Morey and Rouder, 2011](#), p. 411-412). The proof relies on the observation that the replacement of the point-null restriction on the test-relevant parameter, i.e., $\mathcal{H}_0 : \delta = 0$, where $\theta = (\delta, \theta_0)$ as before, yields a peri-null model that defines the same likelihood function as the alternative model. Consequently, the numerator and the denominator of the peri-null Bayes factor $\text{BF}_{10}(Y^n)$ only differ in how the priors are specified.

The inconsistency of peri-null Bayes factors then follows quite directly from Laplace's method ([Laplace, 1774/1986](#)) and consistency of the maximum likelihood estimator (MLE). Both Laplace's method and consistency of the MLE hold under weaker conditions than stated here, namely, for absolute continuous priors (e.g., [van der Vaart, 1998](#), Chapter 10), and regular parametric models (e.g., [van der Vaart, 1998](#), Chapter 7; [Ly et al., 2017](#), Appendix E). These models only need to be one time differentiable with respect to θ in quadratic mean and have non-degenerate Fisher information matrices that are continuous in θ with determinants that are bounded away from zero and infinity. The inconsistency of the peri-null Bayes factor is therefore expected to hold more generally.

We show that under the stronger conditions of [Kass et al. \(1990\)](#), the asymptotic sampling distribution of peri-null Bayes factors can be easily derived. These stronger conditions imply that the model is regular for which we know that the MLE is not only consistent, but also locally asymptotically normal with a variance equal to the inverse observed Fisher information matrix at $\hat{\theta}$ with entries

$$[\hat{I}(\hat{\theta})]^{a,b} = -\frac{1}{n} \sum_{i=1}^n \left(\frac{\partial^2}{\partial \theta_a \partial \theta_b} \log f(Y_i | \theta) \right) \Big|_{\theta=\hat{\theta}}, \quad (6)$$

see for instance [Ly et al. \(2017\)](#) for details.

Theorem 1 (Limit of a peri-null Bayes factor) *Let $Y^n = (Y_1, \dots, Y_n)$ with $Y_i \stackrel{\text{iid}}{\sim} \mathbb{P}_\theta \in \mathcal{P}_\Theta$, where $\mathcal{P}_\Theta$ is an identifiable family of distributions that is Laplace-regular ([Kass et al., 1990](#)). This implies that $\mathcal{P}_\Theta$ admits densities $f(y^n | \theta)$ with respect to the Lebesgue measure that are six times continuously*